

ICC X IGNYTE HACKATHON 2026

ComeBack

The clinician-connected, camera-assisted journey that turns the ICC Return to Play Post-Pregnancy Guidelines into practical support on any phone.

Built by Nourin Zaman and Ashra Rahman Tuba

THE HUMAN PROBLEM

Motherhood should not end a sporting career.

“No player should have to choose between motherhood and representing her country.”

Jay Shah, ICC Chairman

A new framework, but still a document

The ICC published a nine-page Return to Play Post-Pregnancy framework in June 2026. Players and care teams still need a practical way to follow it day by day.

The gap is not another workout app. It is an understandable, measurable, clinician-led return journey.

OUR ANSWER

One calm path from recovery to cricket.

1

Personal roadmap

The ICC 6 Rs become clear next steps with source references.

2

Private camera checks

On-device pose estimates reps, time, asymmetry, depth, and FPPA indicators.

3

Symptom safety gate

Reported symptoms stop approval and direct the player to clinical review.

4

Clinician decision

A clinician reviews metrics and approves or holds the next stage.

AI explains. Camera observes. Only the clinician approves.

Easy to understand without training.

1

See today

Current R stage, one next action, and the guideline source.



2

Move

A framing guide and live cues support a short movement check.



3

Check symptoms

A complete, plain-language check-in decides review urgency.



4

Review

The clinician sees the result and approves or holds progression.

The player experience works offline. When connected, derived metrics sync to the care dashboard. Raw camera media is never saved or uploaded.

Private on the phone, accountable in the cloud.

Player PWA

- ✓ Next.js static export
- ✓ IndexedDB offline data
- ✓ Four UI languages
- ✓ Installable service worker

On-device vision

- ✓ MediaPipe PoseLandmarker
- ✓ Web Worker inference
- ✓ Locally hosted model and WASM
- ✓ No media persistence

Clinical sync

- ✓ Render Express API
- ✓ PostgreSQL metrics store
- ✓ Signed role sessions
- ✓ Server-enforced symptom block

59 unit and API tests, 12 mobile and desktop journeys, 100% deterministic-rules coverage, and zero npm audit findings.

We do not automate medical clearance.

Hard boundaries

- ✓ Clinician approval is mandatory
- ✓ Symptoms block advancement
- ✓ AI cannot alter the stored stage
- ✓ Camera metrics are observations, not diagnoses
- ✓ No camera-media column exists

Evidence boundaries

The ICC document supplies the 6 Rs framework and care-team guidance. Movement screens come from separate Level 4 expert consensus and are presented for clinician review, never as ICC pass thresholds.

Designed for the player most likely to be left out.

1
phone

No wearable or specialist camera is required for the prototype.

4
languages

English, Bengali, Hindi, and Urdu, including RTL layout.

0
video uploads

Camera processing stays in the browser.

6
R stages

A consistent path from Ready through Refine.

Offline-first design supports unstable connectivity and keeps the core journey demonstrable even when the cloud sleeps.

Start small, measure safely, then expand.

1

Clinical review

Have an ICC-aligned sports physician and women's-health physiotherapist review wording, rules, and escalation.

2

Member pilot

Run an 8 to 12 week usability pilot with one Member board, its care team, and a small voluntary cohort.

3

Measure

Track completion, clinician response time, reported clarity, accessibility, and opt-in retention. Do not claim medical efficacy from a prototype pilot.

Pilot success means safer coordination and clearer support, not an automated return-to-play decision.

Built for cricket, adaptable across sport.

ICC value

A configurable Member-board platform can digitise a common framework while allowing local care pathways, languages, legislation, and clinical ownership.

Cross-sport path

The care-team workflow, symptom gates, on-device observations, and clinician review can support other sports. Each sport must supply and validate its own content and rules.

A hosted institutional plan can fund maintenance while the player-facing experience remains affordable and accessible.

TEAM COMEBACK

Help every mother return with a plan, a voice, and a care team.

Nourin Zaman

Player experience, visual design, accessibility, localisation, and demo narrative.

Ashra Rahman Tuba

Evidence, deterministic rules, clinician workflow, backend, and safety validation.

Live prototype: nourinzaman2005-droid.github.io/comeback/